Department of Information Technology

Academic Year 2024 – 2025 (Odd Semester)

Degree, Semester & Branch: V Semester B.Tech. IT
Course Code & Title: CCS334 & Big Data Analytics
Name of the Faculty member (s): Mrs.M.Rethina Kumari

Innovative Practice Description

Unit / Topic: Unit 4 / HadoopStreaming

Course Outcome: CO4

Activity Chosen: Think Pair Share

Justification:

- **Think-Pair-Share activity** enables students to think individually and share their knowledge with peers, facilitating collaborative learning.
- **Hadoop Architecture** is a fundamental concept in Big Data, involving distributed storage and processing of data using HDFS (Hadoop Distributed File System) and MapReduce. A well-structured activity helps students grasp its core components like NameNode, DataNode, JobTracker, TaskTracker, and more. The activity encourages students to critically analyze and understand these concepts individually and collaboratively.

Time Allotted for the Activity: 15 minutes

Details of the Implementation:

- The instructor explained the Hadoop Architecture, including its components and workflow, within 30 minutes. Then, the instructor distributed a list of questions to the students.
- Students individually think about the key components and workflow of Hadoop Architecture (e.g., the roles of NameNode, DataNode, etc.). They formulate their answers based on what they understood.
- Students pair up with a nearby classmate to discuss their answers. They clarify doubts and exchange ideas about the Hadoop components and processes, such as HDFS, MapReduce, and YARN.
- Each pair shares their understanding with the class. The instructor moderates the discussion, highlights critical points, and corrects any misconceptions.

CO – PO / PSO mapping:

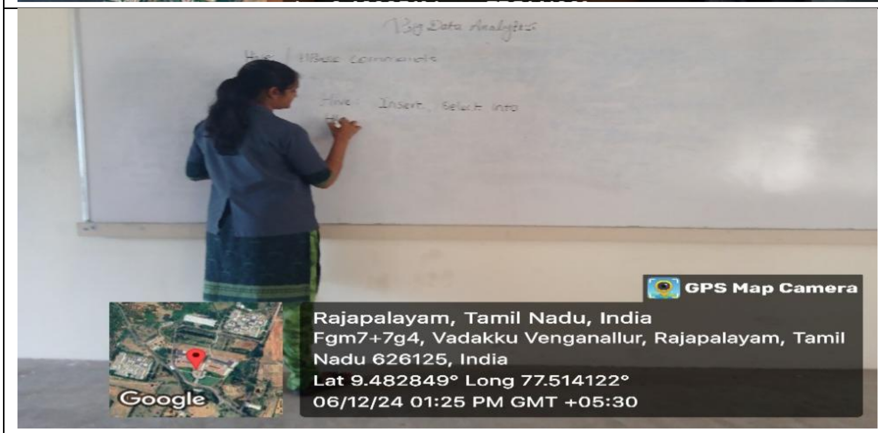
CO	PO1	PO2	PO3	PO9	PO10	PSO1
CO 4	2	1	1	1	1	1

(1 – Low 2 – Moderate 3 – High)

PO / PSO mapped:

Innovative practice	**PO1**	**PO2**	**PO3**	**PO9**	**PO10**	**PSO1**
	2	1	1	1	1	1
Justification for correlation	It includes logical thinking and make the student able to think about problem solving approach	Ableto approach the program by applying what they learnt.	Able to provide solutions for Hadoop architecture related implementation	Students involved in the activity as team, the team work Skill Improved.	Students' communication skill will be improved as they are facing audience and communicating the activity	Provides solutions for Hadoop architecture, including HDFS configuration, MapReduce optimization, and YARN resource management tuning.

- **Images / Screenshot of the practice:**



Glimpse of Think Pair Share Activity

Figure1 shows team discussion of Aarthi .G, Abinaya. S. G, Harshini G and Abinaya S , Figure.2 shows discussion of Aarthi.S with the class members)

❖ **Feedback of practice from students and other stakeholders:**

- Students appreciated that this activity reinforced their understanding of Hadoop Architecture and helped clarify doubts.
- Many found the activity engaging and useful for preparing for assessments.
- Students noted that peer discussions made technical concepts easier to grasp.

❖ **Benefit of the practice:**

- Enhanced clarity on the components and functioning of Hadoop Architecture.
- Students were able to explain concepts more confidently.
- Improved classroom engagement compared to traditional lectures.

❖ **Challenges faced in implementation:**

- Some students struggled to recall the concepts during individual thinking but gained clarity during peer discussions.
- Not all pairs could complete the tasks within the allotted time, requiring additional follow-up.

References:

1. https://www.ritrjpm.ac.in/images/computer-science/TPS-Venkat-GE6075.pdf
2. https://www.readingrockets.org/strategies/think-pair-share
3. https://www.wgu.edu/heyteach/article/how-think-pair-share-activity-can-improve-your-classroom-discussions1704.html
4. https://www.ritrjpm.ac.in/images/B.Tech-IT/20232024/IP/GM_GE3151_U3_Think_pair.pdf

Signature of Faculty Member **HOD**